Probabilistic Matrix Factorization for Movie-Rating Prediction: MAP vs. Variational Inference

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Abstract

We study probabilistic matrix factorization (PMF) as a recommender on a MovieLens subset, framing the model as a graphical model over per-user and per-movie latent vectors and comparing two inference strategies: MAP point estimation and mean-field stochastic variational inference (SVI). Using held-out RMSE we measure how accuracy responds to the latent dimension, the amount of training data, and the inference method. Both PMF variants beat strong mean baselines by roughly 15%, accuracy saturates near a latent dimension of ten without overfitting, the learning curve is still improving at full data, and the Bayesian SVI estimate outperforms MAP on every random split we tried — with its advantage growing sharply as the matrix is made sparser.

1 Introduction

A recommender predicts the ratings a user would give to items they have not rated. We treat this as a missing-entry problem on a sparse user-by-movie rating matrix and study *probabilistic matrix factorization* (PMF) [1]. Each user i is assigned a latent vector $U_i \in \mathbb{R}^K$ and each movie j a vector $V_j \in \mathbb{R}^K$, with zero-mean Gaussian priors; an observed rating is modeled as Gaussian noise about their inner product:

$$U_i \sim \mathcal{N}(0, \sigma_p^2 I_K), \qquad V_j \sim \mathcal{N}(0, \sigma_p^2 I_K),$$
$$r_{ij} \sim \mathcal{N}(U_i^\top V_j,\ \sigma^2).$$

This is a graphical model over thousands of latent variables, and inference amounts to recovering the factors from the observed ratings. We compare two treatments. **MAP** returns the single most probable U, V; with Gaussian priors this is exactly ℓ_2-regularized matrix factorization. **SVI** approximates the full posterior with a mean-field Gaussian over the factors, capturing parameter uncertainty rather than a point estimate. Since matrix factorization itself is standard, the report concentrates on the empirical questions: how RMSE depends on the latent dimension K, on training-set size, and on the inference method.

2 Resources and Code

Data. We use the MovieLens-1M dataset [2] provided on the course Canvas: 1,000,209 ratings from 6,040 users over 3,706 movies on a 1–5 star scale. To obtain a dense matrix suitable for factorization we select the 500 most-rated movies and, among their raters, the 2,000 most-active users. This yields a 2000×500 matrix with 363,568 observed ratings (≈36% filled).
Software. The project is implemented in Python using Pyro [3] (probabilistic programming over PyTorch), with NumPy for data handling and Matplotlib for figures.
Code. We wrote the data-subsetting, model definition, training loop, evaluation, and baselines ourselves (`pmf.py`, `experiments.py`). The PMF model and SVI training follow the structure introduced in HW5; the MAP and SVI guides use Pyro's `AutoDelta` and `AutoNormal` autoguides. No external recommender-system code was used. Source: https://github.com/JonathanKwok777/CS179Project.

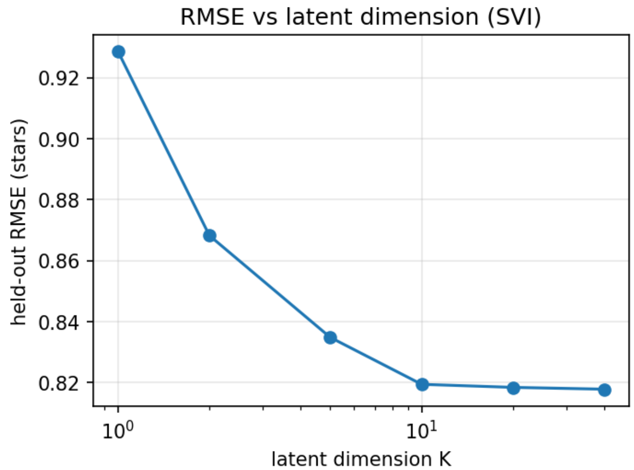

Figure 1: Held-out RMSE vs. latent dimension K (SVI). Error drops sharply then plateaus near $K{\approx}10$; no overfitting appears at $K{=}40$.

3 Evaluation

3.1 Experimental setup

We randomly partition the observed ratings into an 80/20 train/test split (290,854 training, 72,714 held-out test ratings), fixed across all experiments so results are comparable. The model is trained only on the training ratings; we report root-mean-squared error (RMSE) on the held-out ratings, in star units. Ratings are mean-centered before fitting and predictions are de-centered and clipped to $[1, 5]$. Inference uses the Adam optimizer (learning rate 0.01) for 1500 ELBO steps, with unit-scale priors ($\sigma_p = 1$) and unit observation noise ($\sigma = 1$). On a CPU (Apple M-series) a single fit takes ≈90s, and MAP and SVI are comparable in wall-clock time despite the mean-field guide doubling the number of optimized parameters. Mean-centering removes the need to model a global rating offset, and the unit-scale priors supply mild ℓ_2 regularization on the factors. As reference points we report two baselines: predicting the global training mean, and predicting each user's own training mean (falling back to the global mean for any user unseen in training).

3.2 Effect of latent dimension

Figure 1 sweeps the latent dimension K under SVI. RMSE falls steeply from 0.929 at $K{=}1$ to 0.835 at $K{=}5$, then flattens: 0.819 at $K{=}10$ and 0.818 at $K{=}40$. The absence of any upward turn at large K is informative — the Gaussian priors regularize the factors effectively, so the model does not overfit even with forty latent dimensions. Because accuracy saturates around $K{\approx}10$, we fix $K{=}10$ for the remaining experiments as the smallest model that achieves near-best error. We sweep K under SVI only; we probe MAP's overfitting behavior directly in the data dimension in §3.5.

3.3 Effect of training-set size

Holding $K{=}10$, we train on 10% to 100% of the training ratings and evaluate on the same held-out set (Figure 2). RMSE decreases monotonically from 0.946 (29K ratings) to 0.819 (291K ratings). The curve is still descending at full data, indicating the model is data-limited rather than capacity-limited at this scale: additional ratings would be expected to further improve accuracy.

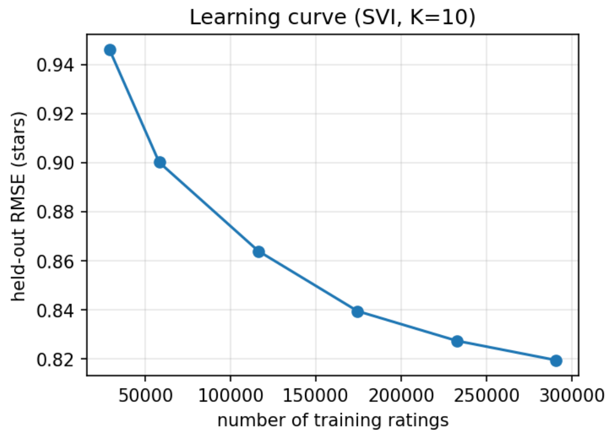

Figure 2: Learning curve (SVI, K=10): held-out RMSE vs. number of training ratings. Error is still falling at full data.

Method	**Held-out RMSE**
Baseline: global mean	1.028 ± 0.003
Baseline: per-user mean	0.968 ± 0.002
PMF, MAP (K=10)	0.828 ± 0.002
PMF, SVI (K=10)	$\mathbf{0.819 \pm 0.001}$

Table 1: Held-out RMSE by method (star units), mean ± std over 5 random train/test splits. Lower is better.

3.4 MAP vs. SVI vs. baselines

Table 1 compares the two inference methods against the baselines at K=10, averaged over five random train/test splits (the sweeps above use a single representative split). Both PMF variants dramatically outperform the mean baselines: the per-user mean attains RMSE 0.968, while PMF reaches 0.819, a ∼15% reduction. The Bayesian SVI posterior mean (0.819 ± 0.001) beats the MAP point estimate (0.828 ± 0.002), and crucially this ordering held on every seed we tried: the ∼0.010 gap is roughly ten times the run-to-run standard deviation, so it reflects a real difference rather than optimization noise. The gap is nonetheless modest, which we attribute to density: at ≈36% observed entries each latent vector is well-identified by abundant data, so the posterior is sharp and its mean lies close to the MAP optimum. We test this directly in §3.5, where the advantage widens sharply as the matrix is made sparser. SVI also yields a posterior standard deviation for every latent coordinate — the explicitly Bayesian quantity the project calls for.

3.5 Sparsity: where the Bayesian treatment pays off

The modest gap at full density invites a direct test of the hypothesis that variational inference should help more when data is scarce. We thin the training ratings to densities of 5%, 10%, 20%, and ∼29% (the natural maximum after the test split), evaluating on the same held-out test set and averaging over five random seeds per point (Figure 3). The gap is far from constant: it grows from +0.010 stars at 29% density to +0.225 at 5%, a roughly twentyfold increase. At 5% density MAP (RMSE 1.139) is in fact *worse than the global-mean baseline* (1.028) — it overfits the few ratings each factor sees — while SVI holds at 0.914. This confirms the

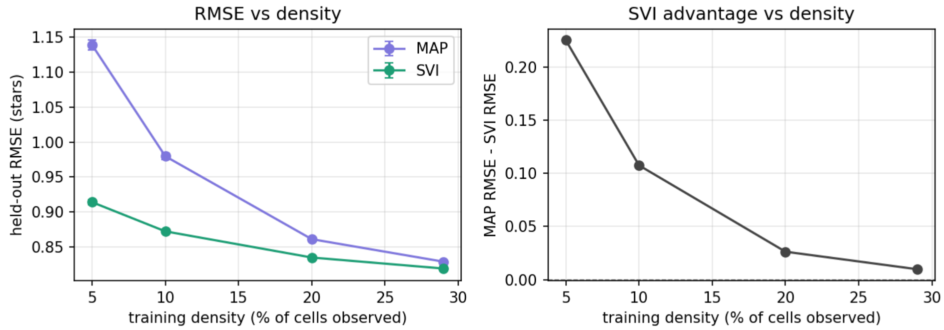

Figure 3: Sparsity sweep (K=10, mean ± std over 5 splits). *Left:* held-out RMSE for MAP and SVI as training density falls. *Right:* the MAP−SVI gap, which grows ∼20× from 29% to 5% density. At the sparsest setting MAP rises above the global-mean baseline (1.028) while SVI does not.

conjectured mechanism: when each latent vector is identified by abundant data the posterior is sharp and its mean sits near the MAP optimum, but as data thins the posterior widens and the point estimate increasingly overfits, so propagating parameter uncertainty becomes decisive. The Bayesian quantity the course calls for is thus not merely present but measurably useful, and most useful exactly where recommenders are hardest.

4 Conclusion

We implemented probabilistic matrix factorization in Pyro and evaluated it as a movie recommender on a dense MovieLens-1M subset via held-out RMSE. Accuracy improves with the latent dimension up to K≈10 and then saturates with no overfitting (the Gaussian priors regularize effectively), and the learning curve is still descending at full data, so the model is data- rather than capacity-limited. Most informatively, mean-field SVI beats MAP by about ten times the run-to-run noise at full density, and this edge grows roughly twentyfold as density falls from 29% to 5% — where MAP overfits below even the global-mean baseline. The variational treatment's advantage is thus small when data is abundant but decisive when it is scarce, exactly the regime where recommenders matter most.

References

[1] A. Mnih and R. Salakhutdinov. Probabilistic Matrix Factorization. *NeurIPS*, 2007.

[2] F. M. Harper and J. A. Konstan. The MovieLens Datasets: History and Context. *ACM TiiS*, 5(4), 2015.

[3] E. Bingham et al. Pyro: Deep Universal Probabilistic Programming. *JMLR*, 20(28), 2019.